

# Quantum resource estimation for minimising energy grid losses

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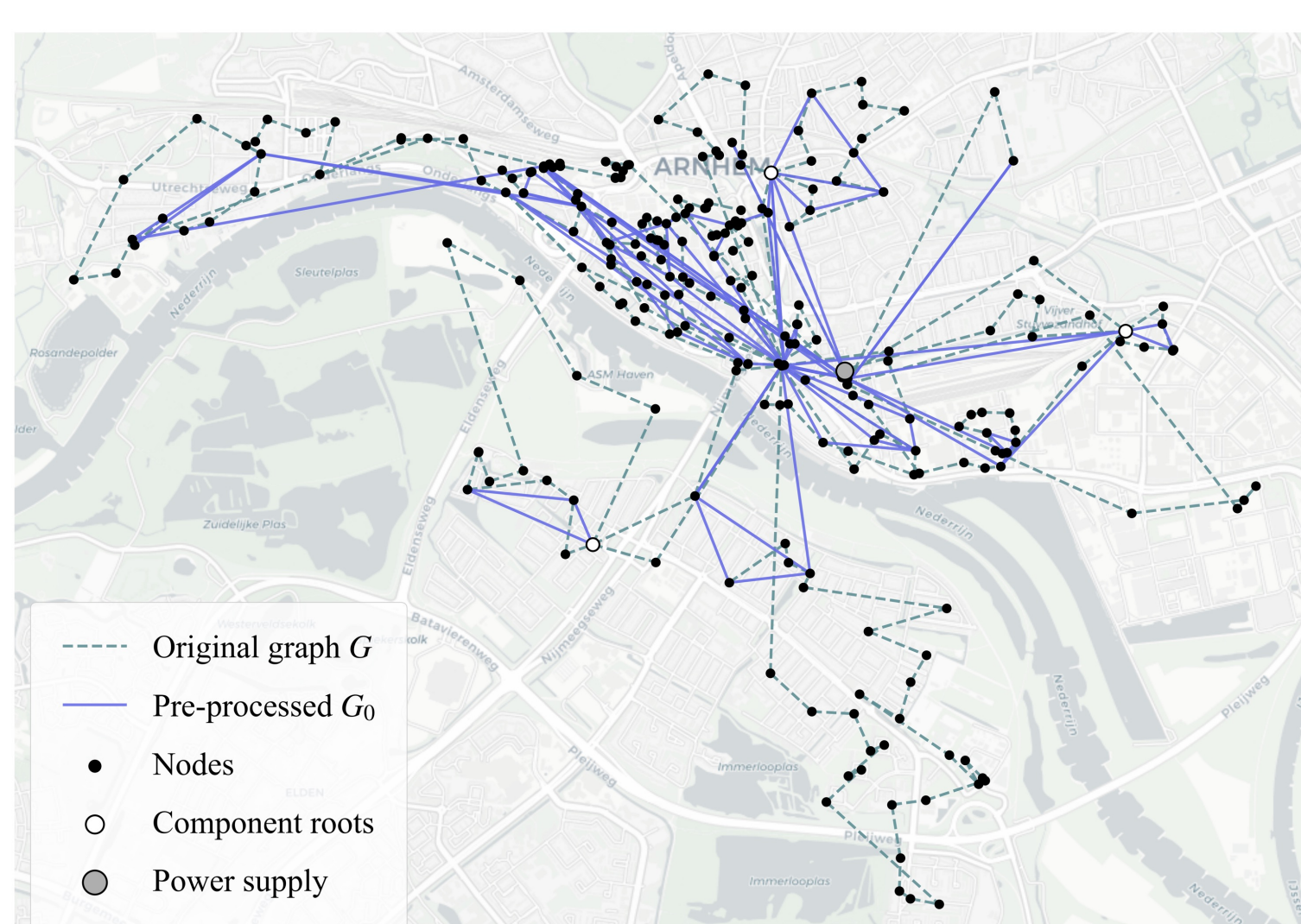
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## Introduction

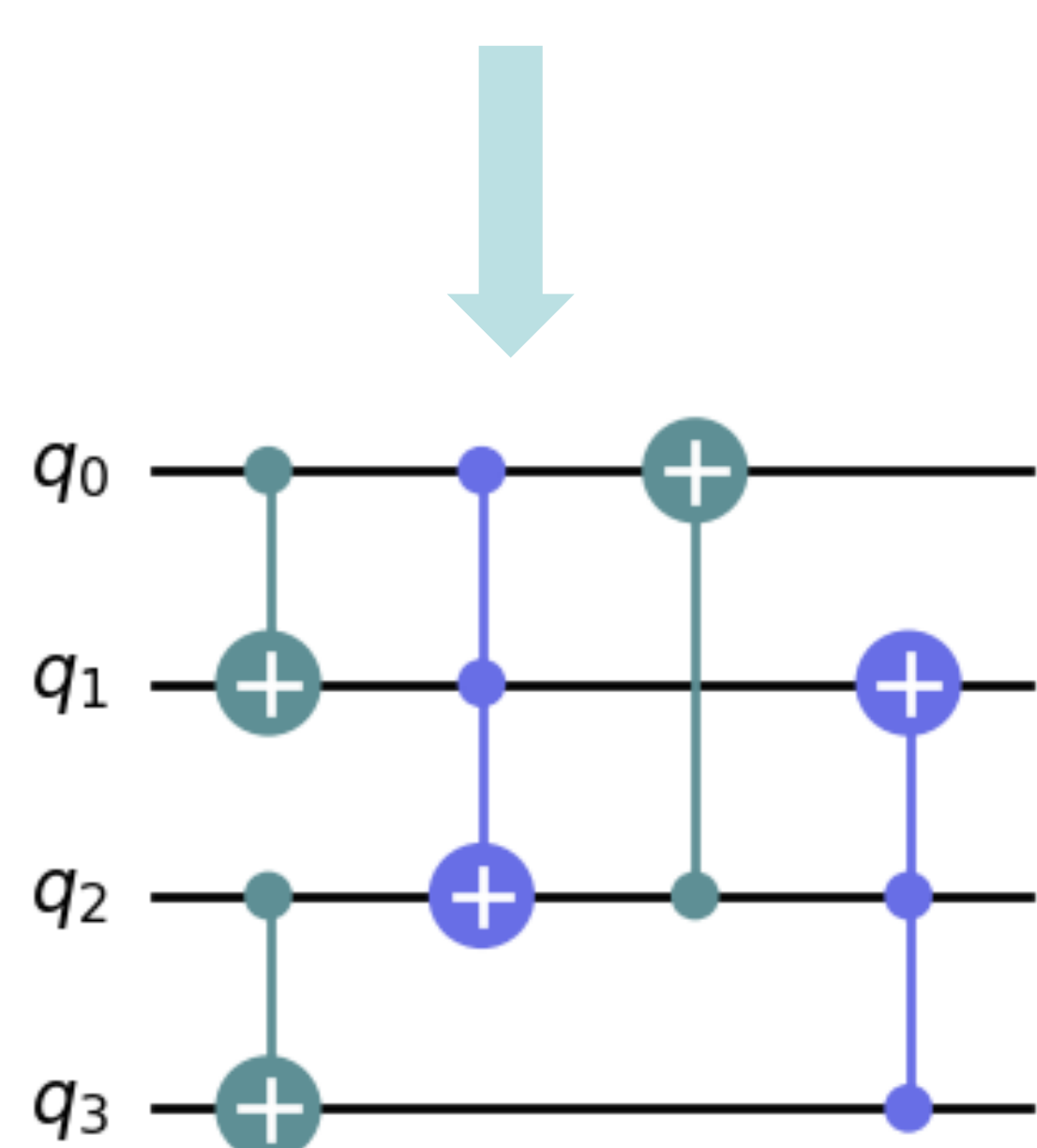
**Distribution network reconfiguration (DNR)** can **reduce power losses** by changing the topology of switchable **electricity grids**. For real medium-voltage (MV) networks, finding the minimum-loss configuration is **computationally hard**. This work investigates whether gate-based **quantum computing** could become feasible for this optimization task. A case-study is done for the real-world MV network in Arnhem.

## Methods



## Data processing

A graph is created from the network data ( $G$ ). With data pre-processing, we obtain  $G_0$ .

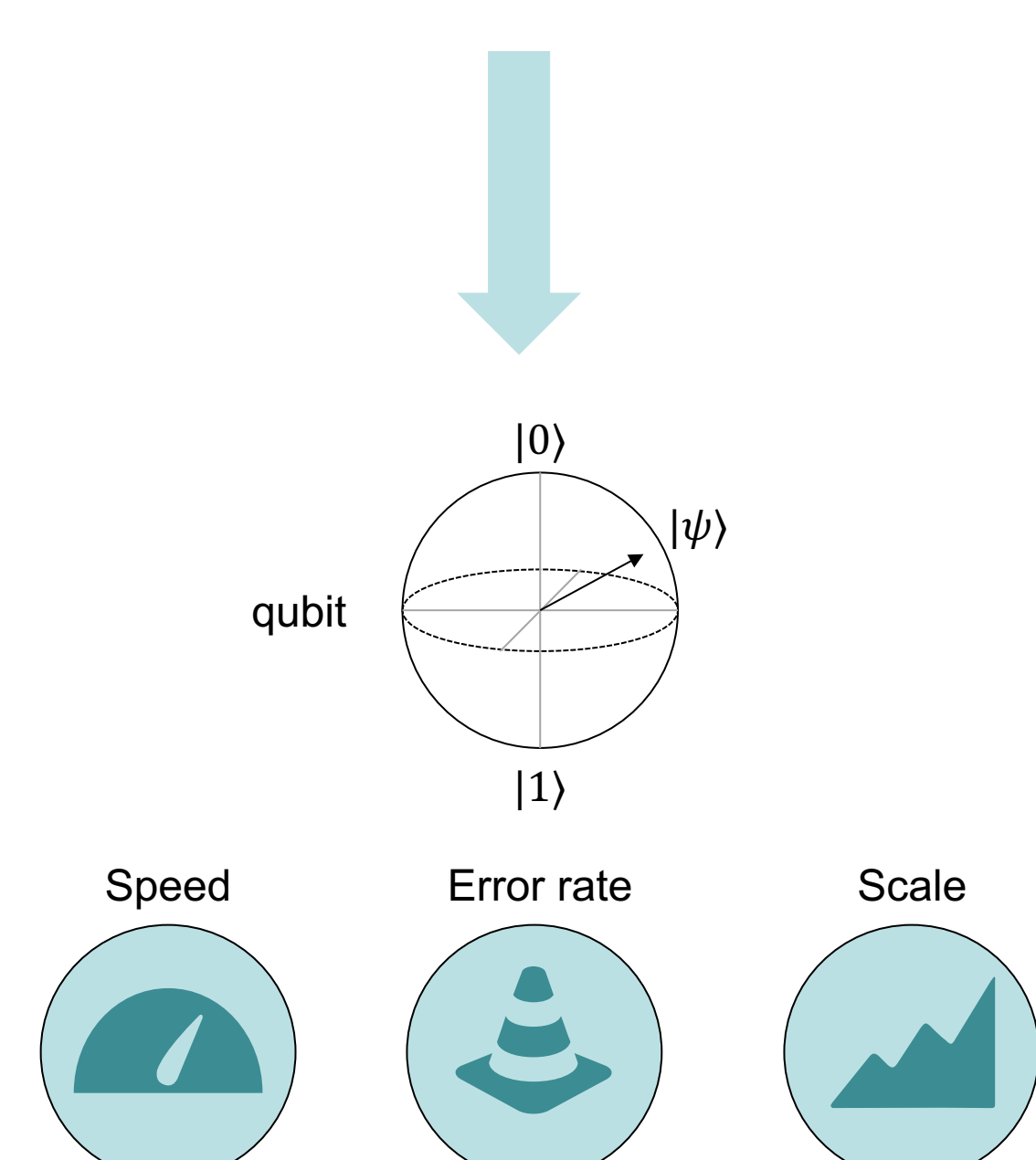


## Quantum algorithm

The DNR problem is formulated as a *higher-order unconstrained binary optimisation (HUBO)* problem.

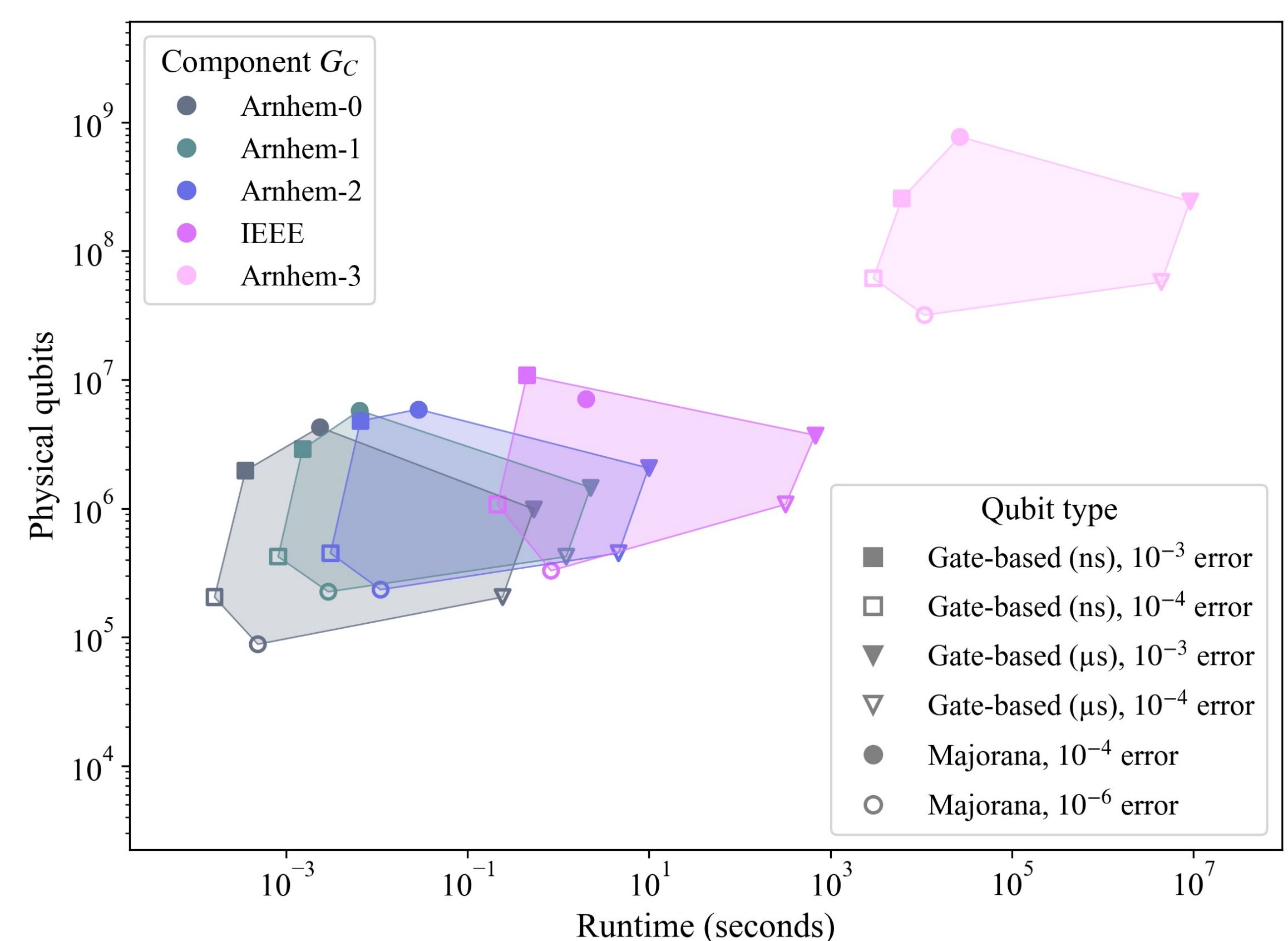
## Quantum resource estimation (QRE)

QRE is performed to gain insights in the necessary quantum hardware to solve the real-world problem.



## Results

The results show that quantum resource requirements increase rapidly with the **size** and **structural complexity** of electricity distribution networks. See figure below.



While the smallest network component required only 42 HUBO **interaction terms**, larger and more connected components produced over  $10^8$  interaction terms, requiring more than 61,172 logical qubits and  $4.94 \times 10^8$  of logical rotation gates. **Runtime** estimates for a single optimisation layer reached the order of  $10^7$  seconds for the largest network.

## Conclusions

- \* Quantum optimisation has potential to aid with the computational complexity of distribution network reconfiguration.
- \* The quantum resource requirements of a real MV network go beyond current hardware roadmaps.
- \* To achieve more realistic values, further algorithmic improvements and research are crucial.

Scan the QR code to read our paper!

